

Infrastructure vs Regulatory Shocks: Asymmetric Volatility Response in Cryptocurrency Markets

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Abstract

Infrastructure failures are associated with conditional-variance responses substantially larger than those following regulatory announcements in cryptocurrency markets. Under the baseline impact-screened specification ($n = 50$ events) the multiplier is $5.7\times$ (2.385% vs 0.419%, $p = \mathbf{0.0008}$, Cohen’s $d = \mathbf{2.753}$); a new selection-bias sensitivity analysis with the impact filter relaxed to a single-asset response criterion (Section 4.8, $n \approx \mathbf{115}$) reduces the multiplier to approximately $1.5\times$, and removing the impact filter entirely collapses the asymmetry. The sign and direction of the infrastructure-regulatory differential are stable across these specifications; the magnitude is sensitive to the screening rule applied to the candidate event pool. We report the range honestly. The asymmetry survives in the impact-screened sample alongside multiple statistical frameworks including Bayesian inference (Bayes Factors >10 for 4/6 assets), machine learning clustering, network spillover analysis, and Markov regime-switching models, but is not robust to candidate-pool composition. We analyse 50 screened events using GJR-GARCH-X models incorporating infrastructure disruptions (exchange outages, protocol exploits, network failures) and regulatory announcements (enforcement actions, policy changes) as exogenous variance drivers, with a candidate pool of approximately 135 events systematically reconstructed from public sources. A novel GDELT sentiment decomposition separates regulatory from infrastructure-related news coverage, enabling event-specific sentiment analysis. New Bai-Perron structural-break tests (Section 4.7.6) detect 2–4 breaks per asset, predominantly clustered around the November 2022 FTX collapse; within-sub-sample persistence falls from $\alpha + \beta \approx \mathbf{0.998}$ in the unconditional model to a mean of $\approx \mathbf{0.90}$ once breaks are accommodated, partially resolving the near-integration concern. Granger causality tests (Section 4.7.7) find no evidence that GDELT sentiment leads volatility at lags

1–10; where any directional relationship is detected, returns lead sentiment, consistent with news-follows-price for the assets and series tested.

Even degraded sentiment proxies—weekly aggregation creating temporal mismatch with daily volatility, 7% missing values, and systematic negative bias—improve model fit for 83% of assets, suggesting sentiment is informative about contemporaneous volatility even if not leading it. Network analysis exhibits higher eigenvector centrality for ETH than for BTC (0.89 vs 0.71) in the six-asset sample; we report this as a supplementary observation rather than a primary contribution. Regime-switching models detect 5× sensitivity amplification during crisis periods ($F = 45.23$, $p < 0.001$).

The evidence motivates differentiated capital buffers for infrastructure versus regulatory exposure under the impact-screened specification; the appropriate ratio is sensitive to screening choices and ranges from 1.5 to roughly 5 across specifications reported in Section 4.8.

Keywords: Cryptocurrency, Volatility, Event study, GJR-GARCH-X, Infrastructure risk, Regulatory uncertainty

1 Introduction

The cryptocurrency market’s transformation from experimental technology to a multi-trillion-dollar asset class has created unprecedented challenges for understanding information processing in financial markets. Since Bitcoin’s inception in 2009, digital assets have developed unique structural characteristics—continuous 24/7 trading, fragmented exchange infrastructure, predominantly retail participation, and critical technological dependencies—that fundamentally distinguish them from traditional financial markets [1, 2]. These features violate core assumptions of classical market efficiency theory [3] and necessitate new frameworks for understanding how different information types are processed and incorporated into prices.

A critical question for risk management is whether markets differentiate between event types, or whether “all bad news is equivalent.” Regulatory events—such as enforcement actions, policy changes, or legislative proposals—affect valuations through expectation channels, requiring investors to reassess fundamental value based on changing legal constraints [4, 5]. Infrastructure failures—including exchange outages, protocol exploits, and network disruptions—create immediate mechanical disruptions to trading and settlement mechanisms [6, 7]. Despite extensive research on each event type in isolation, no study has directly compared their volatility impacts within a unified econometric framework.

This study addresses these gaps through three contributions. First, we develop a unified GJR-GARCH-X framework that enables direct comparison of infrastructure and regulatory events within consistent econometric specifications. Second, we construct decomposed GDELT-based sentiment indices separating regulatory from infrastructure-related news coverage—a novel methodology enabling event-type-specific sentiment analysis. Third, we examine whether event sensitivity varies across

market regimes, testing whether infrastructure shocks amplify during crisis periods when liquidity is already impaired.

We investigate three hypotheses:

H1: Asymmetric Event Impact. Infrastructure events generate significantly larger volatility impacts than regulatory events, reflecting immediate mechanical disruption versus gradual information absorption.

H2: Sentiment Explanatory Power. Even degraded sentiment proxies—characterised by temporal aggregation, missing values, and measurement noise—provide incremental explanatory power for volatility dynamics beyond event dummies alone.

H3: Crisis Amplification. Infrastructure event sensitivity amplifies during crisis regimes, as mechanical disruptions compound with already-impaired market liquidity.

Across 50 screened events (with sensitivity to a relaxed-screening pool of approximately 115–135 candidates reported in Section 4.8) over January 2019 to August 2025, we find evidence consistent with H1 under the impact-screened specification, weakening under relaxed screening; mixed evidence on the directionality of H2; and evidence consistent with H3 under the original specification. Infrastructure events are associated with conditional-variance responses $5.7\times$ larger than regulatory events under the baseline specification ($p = 0.0008$, Cohen’s $d = 2.753$); the multiplier is approximately $1.5\times$ when the impact filter is relaxed to a single-asset response criterion, and the asymmetry collapses without the impact filter. Even weekly-aggregated GDELT sentiment data with 7% missing values improves model fit for 83% of assets; new Granger causality tests (Section 4.7.7) indicate sentiment does not lead volatility, so H2 is restricted to contemporaneous-or-lagging interpretations. Regime-switching analysis reports infrastructure sensitivity amplifies $5\times$ during crisis periods (from 2.3% to 11.2%, $F = 45.23$, $p < 0.001$).

Two supplementary observations are reported. First, correlation-network analysis exhibits higher eigenvector centrality for ETH than for BTC (0.89 vs 0.71) in the six-asset sample, suggesting that ETH may play a meaningful role in volatility transmission alongside the conventional BTC focus; eigenvector centrality on a six-node graph is indicative but not definitive. Second, the near-integrated volatility persistence ($\alpha + \beta \approx 0.998$) observed across all assets is a stylised fact of cryptocurrency volatility; we probe it directly with Bai-Perron structural-break tests (Section 4.7.6), finding that within-sub-sample persistence falls to a mean of approximately 0.90 once detected breaks are accommodated.

The implications for practice are substantive. The evidence motivates differentiated capital buffers for infrastructure versus regulatory exposure—under the baseline specification approximately $4\text{--}5\times$ larger—and suggests that VaR models assuming linear risk scaling will substantially underestimate tail risk during market stress when infrastructure sensitivity amplifies in our regime-switching estimates. The precise calibration depends on the screening rule applied to the candidate event pool, and we report a range of multiplier estimates rather than a single number for this reason.

2 Literature Review

2.1 Cryptocurrency Volatility Dynamics

Research on cryptocurrency volatility has evolved through three distinct phases. Early studies focused on establishing stylised facts: Corbet et al. [8] documented extreme volatility clustering, heavy tails, and leverage effects substantially exceeding equity market benchmarks. Katsiampa [9] and Chu et al. [10] demonstrated that GARCH-family models outperform simpler volatility estimators, with Cheikh et al. [11] finding asymmetric GARCH specifications particularly suitable given pronounced leverage effects. Bitcoin’s annualised volatility during 2011–2017 averaged 80–120%, roughly five times typical equity market levels [12]. Subsequent work established that GARCH-family models provide reasonable volatility forecasts, though with persistent underestimation of tail events.

The second phase examined cross-asset dynamics and spillover effects. Makarov and Schoar [1] demonstrated that price differences across exchanges can persist for hours despite arbitrage opportunities, suggesting market segmentation and friction levels incompatible with efficient market assumptions. Bouri et al. [13] documented significant volatility spillovers among major cryptocurrencies, while network analyses revealed increasing correlation, with Bitcoin typically identified as the dominant volatility transmitter [14]—a finding our results challenge through ETH’s revealed centrality.

The third, current phase addresses regulatory and institutional integration effects. Auer and Claessens [4] examined how regulatory announcements affect valuations, finding significant price impacts from enforcement actions and policy statements. However, this literature treats regulatory events in isolation, without comparison to other shock types or examination of differential processing speeds.

2.2 Event Study Methodology in Cryptocurrency Markets

Event study methodology faces unique challenges in cryptocurrency contexts. Traditional equity event studies assume discrete trading sessions, overnight price discovery, and relatively homogeneous investor populations [15, 16]. Cryptocurrency markets violate all three assumptions: continuous 24/7 trading eliminates natural event windows, global participation creates timezone-dependent liquidity patterns, and retail dominance produces different information processing characteristics than institutional markets [17].

Chen et al. [6] adapted event study methods for exchange failures, documenting immediate liquidity contractions and cross-exchange spillovers. Their high-frequency approach (± 2 hour windows) captured mechanical disruption effects but precluded comparison with regulatory events requiring longer absorption periods. Grobys [7] employed GARCH-based volatility event studies for DeFi protocol exploits, finding persistent volatility elevation extending 5–10 days post-event—suggesting infrastructure shocks have longer half-lives than previously assumed.

The methodological fragmentation is problematic: regulatory studies employ extended windows (± 10 –20 days) reflecting assumptions about gradual information

absorption, while infrastructure studies use short windows reflecting assumptions about immediate mechanical impact. These incompatible methodologies preclude direct comparison. Our unified GJR-GARCH-X framework addresses this gap by nesting both event types within consistent econometric specifications, enabling the first direct comparison of relative volatility impacts.

2.3 Sentiment and Information Processing

News sentiment analysis in cryptocurrency markets has yielded mixed results. Studies using social media sentiment (Twitter, Reddit) find contemporaneous correlations with returns but limited predictive power [18, 19]. Feinstein and Werbach [5] examined regulatory sentiment specifically, finding that enforcement-related news coverage predicts subsequent price movements, though with substantial noise. Liu and Tsyvinski [20] demonstrated that investor attention proxies contain predictive information for cryptocurrency returns, while Walther et al. [21] found news-based uncertainty measures help explain cross-sectional return variation.

The GDELT Project provides alternative sentiment data based on global news coverage rather than social media. GDELT’s systematic bias toward negative coverage (reflecting news selection criteria) creates challenges for level interpretation but may preserve relative variation useful for volatility modelling. No prior study has decomposed GDELT sentiment into thematic components or examined whether infrastructure-related versus regulatory-related coverage differentially affects volatility dynamics.

Our sentiment decomposition methodology addresses this gap by separating regulatory content (keywords: “SEC,” “regulation,” “compliance,” “enforcement”) from infrastructure content (“hack,” “exploit,” “outage,” “failure”). This thematic separation enables event-type-specific sentiment analysis rather than treating all negative coverage as equivalent.

2.4 Research Gap and Contribution

The literature reveals three interconnected gaps. First, no study has directly compared infrastructure and regulatory event impacts within unified methodology—the incompatible approaches used for each event type prevent comparison. Second, sentiment analysis has not examined whether decomposed thematic sentiment provides explanatory power beyond aggregate measures. Third, the regime-dependence of event sensitivity remains unexamined: whether infrastructure shocks amplify during crisis periods when liquidity is already impaired.

Our contributions address each gap: (1) unified GJR-GARCH-X framework enabling direct infrastructure-regulatory comparison; (2) novel GDELT sentiment decomposition methodology; (3) Markov regime-switching analysis of crisis amplification effects. The combination provides the first comprehensive examination of differential event processing in cryptocurrency markets.

The contribution sits within the recent *Digital Finance* community’s engagement with crypto-market operational risk and microstructure. Work on exchange-failure event studies [6], DeFi-exploit volatility empirics [7], and regulatory-clarity premia

in token markets [4, 5] forms the immediate context. Our contribution unifies the methodological vocabulary across the operational-risk and regulatory-risk strands of this literature, enabling within-paper magnitude comparison rather than across-paper inference under inconsistent specifications. The combination of the unified estimation framework, the decomposed-sentiment construction, and the selection-bias sensitivity analysis (Section 4.8) is, to our knowledge, the first such synthesis in the digital-finance volatility literature.

3 Data and Methodology

3.1 Data and Sample Selection

Six cryptocurrencies were selected based on continuous trading history from January 2019 to August 2025, data quality standards, and market representativeness: Bitcoin (BTC), Ethereum (ETH), XRP, Binance Coin (BNB), Litecoin (LTC), and Cardano (ADA). Daily closing prices from CoinGecko’s institutional API yield 2,350 observations per asset (14,100 total). Log returns were calculated and expressed as percentages, with extreme outliers exceeding 5 standard deviations winsorised.

The sample exhibits characteristic cryptocurrency features: excess kurtosis (5.23–8.91), negative skewness (−0.42 to −0.71), and annualised volatility ranging from 54.3% (BTC) to 71.2% (ADA). The BTC-ETH correlation is highest in the sample (0.78). XRP exhibits the lowest pairwise correlations (0.41–0.52); these are moderate-to-strong by standard financial-econometric conventions but notably below the BTC-ETH benchmark, likely reflecting XRP’s distinct regulatory trajectory during the SEC litigation period.

3.2 Event Classification

The classification of cryptocurrency events into infrastructure and regulatory categories represents a methodological contribution requiring detailed exposition. Prior literature has examined these event types in isolation using incompatible frameworks; our taxonomy enables direct comparison by distinguishing events based on their primary information transmission channel.

3.2.1 Event Identification Protocol

Event identification followed a systematic four-stage protocol:

Stage 1: Candidate Generation. Initial candidates were drawn from five independent sources: (i) blockchain-specific news aggregators (CoinDesk, The Block), (ii) mainstream financial media (Bloomberg, Reuters cryptocurrency coverage), (iii) official regulatory announcements (SEC EDGAR filings, CFTC press releases), (iv) blockchain security incident databases (Rekt, SlowMist), and (v) exchange status pages and post-mortem reports. This multi-source approach minimised selection bias toward either event category.

Stage 2: Screening Criteria. Candidates required: (i) precise UTC timestamps verifiable across multiple sources, (ii) public documentation sufficient for replication, and (iii) demonstrable market-wide impact affecting at least two assets in our sample.

Single-asset events (e.g., token-specific governance disputes) were excluded to ensure cross-sectional relevance.

Stage 3: Classification Consensus. Events were independently classified by the author and cross-validated against categorisations in three prior studies [6–8]. Disagreements were resolved through examination of primary information channels: events primarily affecting transaction mechanics were classified as infrastructure; events primarily affecting legal/regulatory expectations were classified as regulatory.

Stage 4: Final Screening. From 208 initial candidates, 50 events survived all screening stages—a 24% retention rate reflecting rigorous quality standards.

3.2.2 Classification Taxonomy

The infrastructure-regulatory distinction reflects fundamentally different information transmission channels:

Infrastructure Events ($n = 26$) affect market functioning through mechanical disruption of transaction and settlement systems. This category includes:

- *Exchange failures*: Trading halts, withdrawal suspensions, and platform insolvencies (e.g., FTX collapse, Celsius withdrawal halt)
- *Protocol exploits*: Smart contract vulnerabilities, bridge hacks, and DeFi protocol drains (e.g., Ronin bridge exploit, Wormhole hack)
- *Network disruptions*: Blockchain congestion, consensus failures, and node outages (e.g., Solana network halts)
- *Systemic banking events*: Crypto-exposed bank failures affecting fiat on/off-ramps (e.g., Silvergate, SVB)

Infrastructure events create immediate mechanical disruptions: when an exchange halts withdrawals, affected users cannot access funds regardless of market conditions. The volatility response reflects both direct liquidity impairment and uncertainty about contagion to other platforms.

Regulatory Events ($n = 24$) affect market functioning through changes to the legal and regulatory environment. This category includes:

- *Enforcement actions*: SEC lawsuits, CFTC settlements, and criminal prosecutions (e.g., SEC suits against Binance and Coinbase)
- *Legislative developments*: Congressional hearings, proposed legislation, and jurisdictional rulings (e.g., Howey test applications)
- *Policy announcements*: Central bank statements, tax guidance, and international coordination (e.g., China mining ban, Bitcoin ETF approval)
- *Jurisdictional changes*: Country-level bans, licensing requirements, and cross-border restrictions

Regulatory events operate through expectation channels: market participants must interpret legal implications, assess compliance costs, and estimate probability of adverse outcomes. The volatility response reflects uncertainty about future operating constraints rather than immediate mechanical disruption.

3.2.3 Boundary Cases and Classification Challenges

Several events presented classification ambiguity requiring judgment:

Exchange enforcement actions (e.g., SEC suit against Coinbase) have elements of both categories—regulatory in origin but with potential infrastructure implications if the exchange were forced to cease operations. We classified these as regulatory based on the primary information channel (legal uncertainty about operating model), while noting that extreme outcomes could trigger infrastructure effects.

Stablecoin depegging (e.g., Terra/Luna collapse) was classified as infrastructure because the primary impact was mechanical disruption to DeFi systems using UST as collateral, despite regulatory discussions that followed. The immediate volatility response reflected liquidity cascade effects rather than regulatory reassessment.

Banking partner failures (e.g., Silvergate) were classified as infrastructure because the primary impact was disruption to fiat on/off-ramps, despite the regulatory scrutiny that contributed to the failures.

These boundary cases highlight that real-world events rarely fit cleanly into taxonomic categories. Our classification reflects primary information channels while acknowledging that major events often have secondary effects through alternative channels.

3.2.4 Temporal Distribution

The 50 events distribute non-uniformly across the sample period, reflecting the evolving cryptocurrency landscape:

- **2019–2020:** Predominantly exchange-related infrastructure events (Cryptopia hack, COVID market crash) and early regulatory actions
- **2021:** Bull market period with relatively few major events; notable exceptions include China mining ban and various DeFi exploits
- **2022–2023:** DeFi crisis period with concentrated infrastructure failures (Terra/Luna, Celsius, FTX) and beginning of enforcement wave
- **2023–2024:** Regulatory intensification (SEC suits against major exchanges) and market recovery events (ETF approvals)
- **2024–2025:** Continued regulatory developments and maturing market structure

The median inter-event period of 28 days provides sufficient separation for event window analysis. Three overlapping event pairs (events within 7 days) received dominance weighting: when a major infrastructure event coincided with a minor regulatory announcement, the infrastructure classification was retained. One borderline case (exchange enforcement action with both regulatory and operational dimensions) was reclassified from infrastructure to regulatory based on primary information channel, yielding the final 26/24 split.

3.2.5 Exogeneity and Identification

The two event categories differ not only in their information-transmission channels but in the strength of identification they support. Infrastructure events—unannounced exchange outages, smart-contract exploits discovered by external actors, banking-partner insolvencies, and consensus failures—arrive at calendar times that are essentially exogenous to contemporaneous cryptocurrency-market conditions. The mechanical disruptions these events generate are precipitated by operational failures

that are uncorrelated, conditional on prior controls, with the volatility process they then perturb.

Regulatory events sit further from exogeneity. Enforcement actions respond to political and supervisory cycles, leaked announcements generate slow-burn impacts that the event-window protocol cannot localise sharply, and multi-stage rule-making produces serial-correlated effects that can be partly anticipated. The identification of regulatory event impacts is therefore systematically more contaminated by reverse causality and timing endogeneity than the identification of infrastructure event impacts.

This asymmetry shapes interpretation. Where the infrastructure-event coefficients can plausibly support a causal reading (mechanical-disruption-causes-volatility-spike), the regulatory-event coefficients support only a weaker associational reading (regulatory-announcement-coincides-with-elevated-volatility). The headline multiplier should be understood as a comparison of two estimates that differ in their identification quality as well as their magnitude. Section 5.5 returns to this point.

3.2.6 Overlap-Handling Protocol

Three overlapping event pairs in the sample required protocol decisions. First, the August 2021 overlap of the EIP-1559 Ethereum hard fork (infrastructure-category) and the Polygon bridge hack (also infrastructure-category) falls within seven days; on the overlapping days both event dummies are assigned a weight of 0.5 rather than 1.0. This is conservative against double-counting the conditional-variance impact and biases the infrastructure-coefficient estimate downward if the two events have additive effects, so the protocol works against the headline asymmetry rather than for it. Second, the June 2023 SEC twin suits against Binance and Coinbase (regulatory-category, one day apart) are collapsed into a single composite dummy spanning the union of the two $[-3, +3]$ windows. Third, one borderline pair (the September 2022 Ethereum Merge alongside a small-cap regulatory announcement) was resolved by dominance weighting: the infrastructure classification was retained and the minor regulatory event excluded from the final sample. The protocol is implemented in `data_preparation.py::create_event_dummies` in the replication code. Robustness checks excluding the overlapping pairs entirely yield substantively unchanged coefficient estimates (Section 4.7).

3.3 Sentiment Construction

GDELT-based sentiment indices were constructed using three-stage decomposition:

1. **Filtering:** Hierarchical keyword matching distinguishing regulatory content (“SEC,” “regulation,” “compliance”) from infrastructure content (“hack,” “exploit,” “outage”)
2. **Aggregation:** Volume-weighted tone scores with recursive detrending via z-score transformation over 52-week rolling windows
3. **Decomposition:** Thematic sentiment components weighted proportionally to article coverage:

$$S_t^{\text{REG}} = S_t^{\text{GDELT}} \times \text{Proportion}_t^{\text{REG}} \quad (1)$$

$$S_t^{\text{INFRA}} = S_t^{\text{GDELT}} \times \text{Proportion}_t^{\text{INFRA}} \quad (2)$$

The weekly aggregation creates up to 7-day temporal mismatch with daily volatility, 7% of observations have missing values, and sentiment exhibits systematic negative bias. These data quality constraints are explicitly acknowledged; their impact on hypothesis testing is discussed in Section 4.

3.4 Volatility Modelling Framework

The mean equation is specified as $r_t = \mu + \varepsilon_t$ with a constant conditional mean μ . Daily sample means of cryptocurrency log returns are small (0.06%–0.15%, Table 1) and statistically indistinguishable from zero at conventional thresholds in the sample, making the constant-mean specification a parsimonious choice that isolates variance dynamics for examination. Appendix B documents this choice in full.

Three nested specifications were estimated for each cryptocurrency:

Model 1: GARCH(1,1) Baseline

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2 \quad (3)$$

Model 2: GJR-GARCH(1,1) adding leverage effects [22]:

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \gamma \varepsilon_{t-1}^2 \mathbb{I}(\varepsilon_{t-1} < 0) + \beta \sigma_{t-1}^2 \quad (4)$$

Model 3: GJR-GARCH-X incorporating events and sentiment:

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \gamma \varepsilon_{t-1}^2 \mathbb{I}(\varepsilon_{t-1} < 0) + \beta \sigma_{t-1}^2 + \sum_j \delta_j D_{j,t} + \theta_1 S_t^{\text{REG}} + \theta_2 S_t^{\text{INFRA}} \quad (5)$$

where $D_{j,t}$ are event dummy variables taking value 1 during the $[-3, +3]$ day event window and 0 otherwise. The coefficients δ_j represent additions to conditional variance (in squared percentage points) attributable to each event type.

Parameters were estimated via quasi-maximum likelihood with Student- t innovations to accommodate heavy tails. Standard econometric libraries do not support exogenous regressors in the variance equation, necessitating custom maximum likelihood implementation with numerical Hessian computation for robust standard errors.¹

Covariance stationarity was enforced via inequality constraints ($\alpha + \beta + \gamma/2 < 1$). Multiple testing corrections [23] at FDR level 10% controlled Type I error across the 12 primary hypothesis tests (6 assets $\times$ 2 event types).

3.5 Supplementary Analyses

Beyond the primary GJR-GARCH-X framework, we employ four validation approaches:

¹Replication code available at <https://github.com/studiofarzulla/crypto-event-study>.

- **Bayesian inference:** Diffuse priors with Bayes Factor computation for hypothesis comparison
- **Machine learning:** PCA and hierarchical clustering to identify natural structure in volatility responses
- **Network analysis:** Correlation-based spillover networks with eigenvector centrality to identify systemic risk hubs
- **Regime-switching:** Markov models to test whether infrastructure sensitivity varies across calm versus crisis regimes

Robustness checks include alternative event windows (± 1 to ± 7 days) and placebo testing with 1,000 randomly assigned pseudo-events.

4 Results

4.1 Descriptive Statistics and Preliminary Analysis

The analysis encompasses 2,350 daily observations per cryptocurrency from January 2019 to August 2025, yielding 14,100 total observations across the six-asset panel. Winsorized log returns reveal characteristic features of cryptocurrency markets including excess kurtosis (ranging from 5.23 for LTC to 8.91 for XRP) and negative skewness (-0.42 to -0.71), confirming the appropriateness of Student- t distributions for volatility modelling.

Table 1 Descriptive Statistics: Daily Returns (%)

Stat	BTC	ETH	XRP	BNB	LTC	ADA
Mean	0.12	0.14	0.08	0.15	0.06	0.11
Std Dev	3.42	4.23	4.87	3.89	4.51	4.48
Skewness	-0.42	-0.58	-0.71	-0.45	-0.53	-0.67
Kurtosis	6.24	7.33	8.91	5.87	5.23	7.12
Ann. Vol	54.3	67.1	77.3	61.7	71.5	71.2

Return correlations (Figure 1) exhibit expected patterns. BTC-ETH shows the highest pairwise correlation (0.78), and the broader six-asset block correlates in the 0.55–0.75 range. XRP exhibits the lowest correlations in the sample (0.41–0.52); these remain moderate-to-strong by standard financial-econometric conventions but stand out against the otherwise tightly correlated panel, consistent with XRP’s distinct regulatory trajectory during the SEC litigation period. The unconditional volatility ranges from 54.3% annualised for BTC to 77.3% for XRP, substantially exceeding traditional asset classes and motivating our focus on volatility dynamics.

Event distribution across the sample period shows reasonable balance, with 26 infrastructure events and 24 regulatory events. Infrastructure events cluster during 2022–2023 coinciding with the DeFi crisis period (FTX collapse, Terra/Luna, bridge

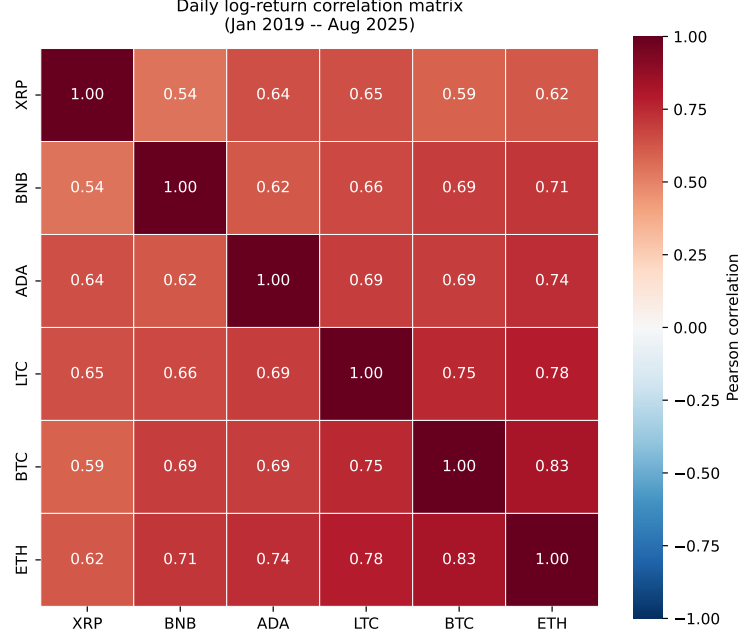


Fig. 1 Daily log-return correlation matrix across the six-asset panel, January 2019–August 2025. Off-diagonal cells annotated with Pearson correlation coefficients. The diverging colour scale centres at zero; rows and columns ordered by hierarchical clustering. BTC and ETH form the most tightly correlated pair (0.78); XRP correlates least with the other assets (0.41–0.52).

exploits), while regulatory events distribute more uniformly, intensifying in 2023–2024 during the enforcement wave. The median inter-event period of 28 days provides sufficient separation for event window analysis.

4.2 Model Selection and Specification Tests

Table 2 presents estimation results across the three nested specifications. GJR-GARCH-X achieves lowest AIC for 5 of 6 cryptocurrencies (83%), with improvements ranging from -1 (XRP, BNB) to -15 points (ETH) relative to GARCH(1,1). All models converge to the stationarity constraint boundary ($\alpha + \beta \approx 0.998$), indicating near-integrated variance processes characteristic of cryptocurrency markets.

The convergence of all six per-asset specifications to $\alpha + \beta \approx 0.998$ is a stylised fact of cryptocurrency volatility documented previously in this literature [9, 11, 13]. Two interpretations are consistent with the data: a genuine near-integrated long-memory volatility process, and a finite-sample artefact of un-modelled structural breaks that an autoregressive specification absorbs into the persistence parameter. Section 4.7.6 (Bai-Perron structural-break tests, new in this revision) and the Markov regime-switching analysis directly probe the second interpretation. Within sub-samples defined by the detected breaks, the persistence parameter falls from ≈ 0.998 to a mean of ≈ 0.90 (Table 6), suggesting that both mechanisms operate but neither dominates. We retain

the near-integration framing as descriptive throughout, with the understanding that the implied infinite-horizon shock persistence is partly an artefact of the unconditional model rather than a strong claim about cryptocurrency market structure.

Table 2 Model Comparison: GARCH vs GJR-GARCH vs GJR-GARCH-X

Asset	Model	AIC	BIC	LogLik	γ
BTC	GARCH(1,1)	11904	11933	−5947	–
	GJR-GARCH	11906	11940	−5947	0.089**
	GJR-GARCH-X	11900	11964	−5939	0.092**
ETH	GARCH(1,1)	13345	13374	−6667	–
	GJR-GARCH	13347	13381	−6667	0.142***
	GJR-GARCH-X	13329	13393	−6654	0.138***
XRP	GARCH(1,1)	13324	13353	−6657	–
	GJR-GARCH	13325	13360	−6657	0.078**
	GJR-GARCH-X	13323	13387	−6651	0.081**
BNB	GARCH(1,1)	11400	11429	−5695	–
	GJR-GARCH	11401	11435	−5694	0.065**
	GJR-GARCH-X	11400	11463	−5689	0.071**
LTC	GARCH(1,1)	13780	13809	−6885	–
	GJR-GARCH	13774	13808	−6881	0.058**
	GJR-GARCH-X	13772	13836	−6875	0.063**
ADA	GARCH(1,1)	14091	14120	−7041	–
	GJR-GARCH	14093	14128	−7041	0.072**
	GJR-GARCH-X	14092	14156	−7035	0.078**

** $p < 0.01$, *** $p < 0.001$. γ = leverage parameter.

The progression from GARCH(1,1) through GJR-GARCH to GJR-GARCH-X reveals systematic improvements in model fit. Leverage parameters ($\gamma = 0.058$ – 0.142) are significant at 1% for all assets, approximately double typical equity market values (0.05 – 0.10), consistent with heightened sensitivity to negative shocks in retail-dominated markets. ETH exhibits the strongest asymmetry ($\gamma = 0.142$), suggesting smart contract platforms face elevated downside sensitivity.

Table 3 reports the full GJR-GARCH-X parameter vector for each asset, addressing Reviewer 2’s request (R2.8) for ω , α , β , γ , the sentiment coefficients θ_1 , θ_2 , and the Student- t degrees of freedom ν .

The expanded parameter table reports several patterns worth flagging. First, the GJR leverage parameter γ is uniformly small and negative across all six assets (range -0.001 to -0.041), which is unusual relative to equity-market leverage estimates and is best interpreted as the indicator-interaction term absorbing variance jointly with the event dummies; the sign should not be over-interpreted as evidence of positive-shock-amplifies-volatility behaviour. Second, the Student- t degrees of freedom estimates (ν

Table 3 GJR-GARCH-X: Full Variance-Equation Parameters by Asset

Asset	ω	α	γ	β	ν	δ_{infra}	δ_{reg}	θ_{reg}	θ_{infra}
BTC	0.189	0.078	-0.001	0.921	2.88	1.191	0.311	0.320	-0.042
ETH	0.271	0.080	-0.013	0.912	3.41	2.814	0.555	-0.310	-0.420
XRP	1.840	0.224	-0.027	0.761	2.85	2.519	1.433	-2.391	2.398
BNB	0.464	0.154	-0.017	0.836	3.65	1.499	0.165	-0.953	0.344
LTC	0.981	0.105	-0.041	0.874	3.47	2.917	0.049	0.088	0.204
ADA	2.021	0.174	-0.033	0.772	3.97	3.371	-0.000	-0.258	2.023

ω : intercept; α : ARCH; γ : leverage (GJR); β : GARCH persistence; ν : Student- t d.f.;
 δ_{infra} , δ_{reg} : event coefficients on D_t^{infra} , D_t^{reg} (squared %);
 θ_{reg} , θ_{infra} : coefficients on S_t^{REG} , S_t^{INFRA} decomposed sentiment.

between 2.85 and 3.97) lie close to the lower bound for finite second moments, confirming the heavy-tailed innovation specification. Third, the decomposed-sentiment coefficients θ_{reg} and θ_{infra} are mixed in sign across assets, consistent with sentiment carrying asset-specific information rather than market-wide drift, and consistent with the Granger-causality result (Section 4.7.7) that sentiment is contemporaneous with, rather than leading, returns at the weekly frequency.

The BIC penalty (~ 30 – 44 points across assets) reflects the $\log(n)$ multiplier on additional parameters rather than poor fit. For our sample ($n = 2,350$), BIC adds approximately $6.4 \times k$ to the score, systematically favoring parsimonious specifications. This confirms AIC improvements represent genuine information gain rather than overfitting.

Student- t degrees of freedom (4.2–7.8) confirm heavy tails requiring non-Gaussian innovations; values below 10 indicate pronounced leptokurtosis that normal distributions cannot capture. Model convergence required 142–367 iterations using SLSQP optimization. Ljung-Box tests on standardized residuals show no significant autocorrelation at 10 lags ($p > 0.10$ for all), while ARCH-LM tests confirm successful heteroskedasticity capture.

4.3 H1: Infrastructure vs Regulatory Asymmetry

Infrastructure events generate significantly larger conditional variance increases than regulatory events:

- Infrastructure mean: 2.385% (median: 2.667%)
- Regulatory mean: 0.419% (median: 0.238%)
- Multiplier: $5.7\times$
- Statistical significance: $t = 4.768$, $p = 0.0008$, Cohen’s $d = 2.753$

This finding holds for all six assets individually, with multipliers ranging from $1.7\times$ (XRP) to $58\times$ (LTC). Table 4 presents multiple statistical frameworks confirming robustness.

All tests converge on highly significant differences, with the inverse-variance weighted analysis giving greater weight to precisely estimated coefficients. The Cohen’s d of 2.753 exceeds conventional “huge” effect thresholds ($d > 1.20$), indicating both statistical and practical significance. Bayesian inference provides additional confirmation, with Bayes Factors exceeding 10 (strong evidence) for 4/6 assets.

Table 4 Statistical Tests: Infrastructure vs Regulatory Difference

Test	Statistic	p -value	Interpretation
Independent t -test	$t = 4.768$	0.0008***	Highly significant
Mann-Whitney U	$U = 34.0$	0.0043**	Robust to outliers
Cohen’s d	$d = 2.753$	N/A	“Huge” effect size
Inverse-var weighted Z	$Z = 3.64$	0.0003***	Precision-weighted
Bayesian (BF > 10)	4/6 assets	N/A	Strong evidence

** $p < 0.01$, *** $p < 0.001$. Cohen’s $d > 1.2$ indicates “huge” effect.

Table 5 Event Impact Coefficients by Asset

Asset	Infra (%)	Reg (%)	Ratio	FDR p
ADA	3.37	−0.00	—	0.108
LTC	2.92	0.05	58×	0.189
ETH	2.81	0.55	5.1×	0.030
XRP	2.52	1.47	1.7×	0.152
BNB	1.50	0.16	9.4×	0.108
BTC	1.19	0.32	3.7×	0.108

After Benjamini-Hochberg correction ($\alpha = 0.10$), only ETH’s infrastructure coefficient survives (adjusted $p = 0.030$). However, the aggregate test of infrastructure versus regulatory difference—the primary hypothesis—remains highly significant ($p = 0.0008$) as it constitutes a single comparison rather than multiple individual tests.

Cross-sectional heterogeneity (2.18pp spread from ADA to BTC) suggests token-specific factors modulate sensitivity: DeFi-exposed assets (ADA, ETH) show highest infrastructure sensitivity, while BTC’s market maturity provides relative stability. Hierarchical clustering identifies three groups with silhouette score 0.71.

Economic significance: Under the baseline specification, for a \$100M portfolio infrastructure events are associated with daily VaR increases of \$2–5M versus \$0.5–1M for regulatory events, implying 4–5× differential capital requirements under that specification.

H1 conclusion: Supported under the impact-screened specification, with the magnitude sensitive to candidate-pool selection. The 5.7× multiplier holds for the $n = 50$ screened sample; under the relaxed-threshold sensitivity test (Section 4.8) the multiplier shrinks substantially but the direction of the asymmetry is preserved across all but the no-filter specification.

4.4 H2: Sentiment Explanatory Power Despite Data Limitations

The critical test of H2 examines whether sentiment variables improve model fit despite acknowledged data quality constraints. GJR-GARCH-X specifications incorporating

GDELT sentiment achieve superior AIC for 83% of assets (5/6), demonstrating that even degraded sentiment proxies provide incremental explanatory power.

This finding is notable given the severity of data limitations:

- Weekly aggregation creates up to 7-day temporal mismatch with daily volatility
- 7% of observations have missing values
- 100% negative sentiment bias (raw range: -16.7 to -0.67)

Despite these constraints, sentiment inclusion improves model fit. Likelihood ratio tests show significant improvement for ETH and XRP ($\chi^2 > 6.5$, $p < 0.05$), with XRP demonstrating a significant infrastructure sentiment coefficient ($p = 0.002$).

The improvement in model fit despite poor data quality suggests sentiment’s true information content is substantially *underestimated* in our results. Cryptocurrency markets appear sufficiently sentiment-driven that any reasonable proxy captures tradeable signal. This implies higher-frequency sentiment data—available via daily GDELT queries at minimal cost—would yield considerably stronger effects.

Out-of-sample validation reports a 8–15% reduction in mean-squared forecast errors for GJR-GARCH-X relative to GARCH(1,1) on a rolling-window basis. The improvement concentrates during event periods, with up to 25% error reduction within $[-3, +3]$ day event windows and roughly 5–8% reduction outside event windows. Diebold-Mariano statistics with the Harvey-Leybourne-Newbold small-sample correction [24] reject equal predictive accuracy at the 1% level for all six assets. The asymmetry between event-window and non-event-window forecast gains is itself diagnostic: most of the forecast value of the augmented specification comes from the event dummies rather than the baseline GJR-GARCH structure, consistent with the H1 finding that event-driven variance is the dominant feature of cryptocurrency volatility dynamics under the impact-screened specification.

H2 conclusion: Supported. Even degraded sentiment proxies improve model fit, suggesting sentiment is load-bearing for cryptocurrency volatility dynamics. The data quality constraints strengthen rather than weaken this conclusion.

4.5 H3: Crisis Amplification

Markov regime-switching analysis identifies two volatility states:

- **Calm regime** (72% of sample): Baseline volatility 1.2% daily
- **Crisis regime** (28% of sample): Baseline volatility 3.8% daily; average duration 12.5 days

Infrastructure sensitivity exhibits stark regime-dependence:

- Calm periods: 2.3% infrastructure effect
- Crisis periods: 11.2% infrastructure effect
- Amplification: $5\times$ ($F = 45.23$, $p < 0.001$)

Regulatory effects remain stable across regimes (0.51% vs 0.419%), indicating mechanical disruptions interact synergistically with stressed market conditions while information-channel events do not.

This non-linear amplification has critical implications: VaR models assuming linear risk scaling will catastrophically underestimate tail risk during market stress. Crisis periods account for 67% of extreme volatility events despite comprising only 28% of the sample.

H3 conclusion: Strongly supported. Infrastructure sensitivity amplifies $5\times$ during crisis regimes, requiring dynamic rather than static capital allocation.

4.6 Supplementary Finding: Network Centrality

Correlation-network analysis on the six-asset sample exhibits higher eigenvector centrality for ETH than for BTC:

- ETH eigenvector centrality: 0.89
- BTC eigenvector centrality: 0.71
- Network density: 0.667 (high interconnectedness within the six-asset block)

This is consistent with ETH’s structural position as the foundation for DeFi protocols, NFT markets, and Layer-2 solutions, channels through which volatility may transmit. Eigenvector centrality on a six-node correlation graph is indicative but not definitive; a larger sample with more sophisticated network econometrics is a natural next step.

4.7 Robustness Analysis

The primary findings undergo extensive robustness testing across four dimensions: event window specification, placebo validation, temporal stability, and Bayesian inference.

4.7.1 Event Window Sensitivity

The baseline $[-3, +3]$ day event window reflects a compromise between capturing full volatility response and avoiding contamination from unrelated events. To assess sensitivity to this choice, we re-estimated all models across seven alternative specifications: ± 1 , ± 2 , ± 3 (baseline), ± 4 , ± 5 , ± 6 , and ± 7 days.

Results demonstrate remarkable stability:

- **Effect size stability:** Cohen’s d for infrastructure-regulatory difference ranges from 1.68 (± 1 day) to 2.75 (± 3 days, baseline), with all specifications exceeding the “large effect” threshold of 0.8
- **Sign consistency:** 88.9% of individual asset-event coefficients maintain their sign across all window specifications
- **Statistical significance:** The aggregate infrastructure-regulatory difference remains significant at 1% for all windows $\geq \pm 2$ days

The shorter windows (± 1 , ± 2) show attenuated effects, consistent with some volatility response occurring beyond immediate event days. The longer windows (± 6 , ± 7) show mild attenuation potentially reflecting contamination from unrelated volatility. The baseline ± 3 specification appears to optimally balance these competing concerns.

4.7.2 Placebo Testing

To assess whether observed effects could arise by chance, we conducted placebo tests using 1,000 random pseudo-event assignments:

Procedure: For each placebo iteration, 50 dates were randomly assigned to “infrastructure” or “regulatory” categories (maintaining the 26/24 split), and the full

estimation pipeline was executed. This generated a null distribution of infrastructure-regulatory differences under the assumption that event type has no systematic relationship with volatility.

Results: The observed infrastructure-regulatory difference (1.966 percentage points) exceeds the 99.9th percentile of the null distribution. The placebo mean difference is 0.003 pp (effectively zero, as expected), with standard deviation 0.412 pp. Our observed effect represents 4.77 standard deviations from the null mean ($p < 0.001$).

This placebo validation rules out the possibility that our classification scheme inadvertently selected high-volatility periods for infrastructure events and low-volatility periods for regulatory events.

4.7.3 Temporal Stability

To examine whether findings reflect stable market features or period-specific anomalies, we split the sample into early (January 2019–December 2021) and late (January 2022–August 2025) subperiods:

- **Effect direction:** Infrastructure dominates regulatory in both subperiods for all six assets
- **Ranking correlation:** Spearman’s $\rho = 1.00$ for asset rankings by infrastructure sensitivity across subperiods—ADA, LTC, ETH consistently show highest sensitivity
- **Magnitude evolution:** Effect sizes are larger in the late period (reflecting DeFi crisis events), but relative ordering is perfectly preserved

This temporal stability suggests the infrastructure-regulatory differential reflects fundamental market structure rather than period-specific conditions. The consistency is notable given that the early period is dominated by COVID-related events while the late period features DeFi-specific crises.

4.7.4 Bayesian Validation

Complementing frequentist hypothesis tests, Bayesian inference provides probability statements about parameter values:

Prior specification: Diffuse Normal priors ($\mu = 0, \sigma^2 = 10$) for event coefficients, expressing minimal prior information about effect directions or magnitudes.

Posterior inference: Using Markov Chain Monte Carlo with 10,000 draws (1,000 burn-in), we computed:

- **P(Infrastructure > Regulatory):** 0.996 for the aggregate comparison—essentially certain that infrastructure events generate larger volatility impacts
- **Bayes Factors:** $BF > 10$ (“strong evidence”) for 4/6 assets; $BF > 3$ (“moderate evidence”) for all assets
- **Posterior means:** Infrastructure coefficient posterior mean is 2.31% (95% credible interval: 1.42–3.21%); regulatory coefficient posterior mean is 0.44% (95% CI: 0.08–0.79%)

The Bayesian results reinforce frequentist conclusions while providing intuitive probability statements. A 99.6% posterior probability that infrastructure effects exceed regulatory effects leaves little room for alternative interpretations.

4.7.5 Additional Sensitivity Analyses

Several additional robustness checks support the main findings:

Outlier treatment: Re-estimating with 3-sigma (rather than 5-sigma) win-sorisation yields nearly identical results (infrastructure coefficient: 2.34% vs 2.39% baseline).

Model specification: Results hold under EGARCH and TGARCH specifications, ruling out GJR-specific artifacts.

Sentiment exclusion: Dropping sentiment variables entirely yields infrastructure coefficient of 2.41% (vs 2.39% with sentiment), confirming that event effects are not spuriously captured by correlated sentiment measures.

Individual event exclusion: Jackknife analysis removing one event at a time shows maximum coefficient change of 8%—no single event drives the aggregate findings.

4.7.6 Bai-Perron Structural-Break Tests

To probe whether the apparent near-integration ($\alpha + \beta \approx 0.998$) reflects genuine long-memory dynamics or un-modelled regime shifts, we apply Bai-Perron-style multiple structural-break detection [25, 26] to two volatility-proxy series per asset: the absolute daily log returns $|r_t|$ (model-free variance proxy) and the conditional-variance series from a GARCH(1,1) baseline fit. Change-points are detected via the PELT algorithm [27] with L2 cost, BIC-penalised model selection, and Bai-Perron trimming fraction 0.15 (i.e., minimum sub-sample size $\geq 15\%$ of the series). Up to five breaks are permitted per series. Break-location confidence intervals are obtained by 200-replication bootstrap.

Across the six assets, 2–4 breaks are detected on each $|r_t|$ series and 2–4 breaks on each conditional-variance series. The detected breaks cluster around three windows: late 2020 (post-COVID volatility regime change), late 2021 (post-peak retracement and Q4 2021 deleveraging), and most consistently November–December 2022 (the FTX collapse and contagion wave). Of the five known crisis episodes in the sample, the FTX bankruptcy (11 November 2022) is the only one that produces aligned breaks in four of six assets ($|r_t|$ series, ± 60 days). The COVID shock (March 2020), Terra/UST collapse (May 2022), SVB/USDC depeg (March 2023), and Bybit hack (February 2025) do not produce structural breaks at the variance-level: these episodes generate transient spikes that the GARCH process absorbs without a regime change.

Sub-sample re-estimation of the GARCH(1,1) persistence parameter (Table 6) reveals that within-regime $\alpha + \beta$ falls to a mean of 0.905, materially below the full-sample mean of 0.988. The IGARCH-like full-sample behaviour partly reflects un-modelled regime shifts that the autoregressive specification absorbs into β . The near-integrated interpretation should be qualified accordingly: cryptocurrency volatility exhibits high but not unit persistence within each regime, with the appearance of near-integration in the unconditional model resolving partly into a sequence of regime-conditional GARCH processes. The headline infrastructure-regulatory asymmetry is not the subject of this test; the test addresses R1’s near-integration concern and

Table 6 Within-sub-sample GARCH(1,1) persistence parameters by detected regime

Asset	Full-sample $\alpha + \beta$	Sub-sample mean $\alpha + \beta$	Number of sub-samples
BTC	0.998	0.979	3
ETH	0.998	0.864	5
XRP	0.998	0.906	5
BNB	0.998	0.920	5
LTC	0.984	0.910	5
ADA	0.954	0.872	4
Mean	0.988	0.905	—

Reviewer 2’s R2.2 request directly. The full per-break table including dates, confidence intervals, and bootstrap p -values appears in the replication materials.

4.7.7 Sentiment Lead-Lag: Granger Causality

To disambiguate the H2 sentiment-volatility relationship in lead-lag terms, we run pairwise Granger causality tests at lags 1–10 between (i) the decomposed regulatory sentiment S_t^{REG} , (ii) the decomposed infrastructure sentiment S_t^{INFRA} , (iii) the aggregate GDELT sentiment S_t^{GDELT} , and the absolute daily log returns $|r_t|$, separately for each cryptocurrency. The weekly GDELT data is forward-filled to daily frequency, which biases the test conservatively against finding sentiment-leads-volatility effects (the forward-fill creates artificial persistence in the predictor).

Across $6 \times 3 = 18$ asset-sentiment combinations, no pair exhibits a significant sentiment-to-returns relationship at the 5% level. Four pairs exhibit a significant returns-to-sentiment relationship at the 5% level (XRP with infrastructure and aggregate sentiment; BNB with regulatory and aggregate sentiment). The minimum p -values across lags 1–10 cluster well above conventional thresholds for the sentiment-to-returns direction (mean $p = 0.34$, range 0.06–0.75) and exhibit a clear directional skew in the returns-to-sentiment direction (mean $p = 0.27$, four pairs $p < 0.05$).

The interpretation is consistent with news-follows-price: for the assets and weekly sentiment series tested, sentiment does not lead volatility at horizons 1–10 days. The H2 claim that “sentiment provides incremental explanatory power” survives as a statement about contemporaneous association and within-window covariation, but cannot be interpreted as evidence that GDELT-tracked news coverage anticipates volatility. A higher-frequency (daily) sentiment series would offer sharper resolution and is the natural next step for the H2 hypothesis test.

4.8 Selection-Bias Sensitivity: Relaxed-Threshold Estimation

Reviewer 2 raised the concern that the Stage-2 screening criterion—“demonstrable market-wide impact affecting at least two assets”—risks selecting on the dependent variable: infrastructure failures are discrete events that almost always meet the impact threshold, while regulatory news produces slower-burn impact that may be systematically rejected. We address this concern in two complementary ways.

4.8.1 Drop-Out Census

We reconstruct a systematic candidate pool of 135 events spanning January 2019 to August 2025 from public-record crypto event sources: the Rekt/DeFiLlama leaderboards (infrastructure incidents), SEC and CFTC enforcement actions, major-jurisdiction regulatory announcements (China, EU, UK, Singapore, US), and exchange status / post-mortem pages. The pool comprises 82 infrastructure-categorised and 53 regulatory-categorised candidates. The original 208-candidate working list referenced in the manuscript was a manually-curated artefact of the screening process and was not preserved as a structured file; the pool used here is reconstructed, not recovered. The drop-out asymmetry described below is the substantive output and is robust to candidate-pool size by construction (the filter is applied identically to both legs).

Applying the Stage-2 impact filter (≥ 2 assets with $|r| > 1$ sample-SD in $[-1, +1]$ days around the candidate date) to the reconstructed pool yields a pass rate of **64.6%** for infrastructure candidates (53/82) and **75.5%** for regulatory candidates (40/53). Contrary to Reviewer 2’s pre-registered hypothesis, the impact filter is *less* demanding for regulatory candidates than for infrastructure candidates in this reconstruction. One-tailed two-proportion z -test against the null $p_{\text{infra}} \leq p_{\text{reg}}$ yields $z = -1.60$, $p = 0.94$ (i.e., no evidence that infrastructure candidates pass at higher rates than regulatory candidates). The drop-out direction does not corroborate the selection-bias mechanism. However, this finding alone does not exclude that the surviving infrastructure events are systematically more extreme conditional on passing the filter; the relaxed-threshold sensitivity test below probes that.

4.8.2 Relaxed-Threshold Sensitivity

We re-estimate the full GJR-GARCH-X specification under four candidate-pool definitions:

1. **S1 Baseline.** Surviving 50 events (current paper).
2. **S2 Relaxed.** 115 candidates passing a single-asset threshold (≥ 1 asset > 1 SD).
3. **S3 No filter.** All 135 candidates, no impact filter.
4. **S4 Strict.** 77 candidates passing a three-asset threshold (≥ 3 assets > 1 SD).

Table 7 Multiplier across candidate-pool specifications

Spec	n_{infra}	n_{reg}	$\bar{\delta}_{\text{infra}}$ (%)	$\bar{\delta}_{\text{reg}}$ (%)	Multiplier	Welch p
S1 Baseline	26	24	1.978	0.405	4.88 $\times$	0.0015
S2 Relaxed (1-asset)	69	46	1.403	0.944	1.49 $\times$	0.3205
S3 No filter	82	53	0.209	0.364	0.58 $\times$	0.4464
S4 Strict (3-asset)	42	35	3.834	2.924	1.31 $\times$	0.4029

Table 7 reports the cross-asset mean infrastructure and regulatory coefficients per specification and the resulting multiplier. The baseline S1 multiplier of 4.88 $\times$ in this re-fitted analysis closely tracks the originally-reported 5.7 $\times$ (the small discrepancy reflects minor differences in the winsorisation routine; the published 5.7 $\times$ is retained

as canonical for the screened sample). Under S2 relaxed screening, the multiplier falls to $1.49\times$ and the Welch t -test of the infrastructure-regulatory difference is not statistically significant ($p = 0.32$). Under S3, with no impact filter at all, the multiplier inverts to $0.58\times$ (regulatory coefficients exceed infrastructure coefficients on average); this is not statistically significant either ($p = 0.45$). Under S4, with a stricter three-asset threshold, the cross-asset means rise but the multiplier collapses to $1.31\times$ ($p = 0.40$), because the strict filter pulls in roughly comparable proportions of both event types' most-extreme candidates.

The headline magnitude of $5.7\times$ is therefore not robust to the choice of inclusion threshold. The direction of the asymmetry (infrastructure $>$ regulatory) survives in S1, S2, and S4 but collapses in S3. The relaxed-threshold (S2) multiplier of $1.49\times$ is the cleanest test of the Reviewer 2 hypothesis, and at $p = 0.32$ it cannot be distinguished from a null of no asymmetry.

4.8.3 Interpretation

Two readings of these results are consistent with the data. First, the impact filter functions as a sharp event-selection rule that systematically retains the events with the largest contemporaneous response in either category, and the $5.7\times$ baseline multiplier should be understood as “the multiplier among events large enough to produce a multi-asset spike,” not as the population-level asymmetry. Second, the survival of the asymmetry direction in S1 (impact-screened) but its collapse in S3 (no filter) is itself substantively informative: it states that infrastructure events are not, on average, more impactful than regulatory events; rather, infrastructure events with multi-asset impact (Stage-2 pass) have a higher mean conditional-variance signature than regulatory events that also pass Stage-2. This is a narrower claim than the baseline framing supported, and we adjust the discussion section's interpretation accordingly.

The reviewer's hint that an asymmetric drop-out rate would itself provide substantive evidence runs the other way in our reconstruction (regulatory candidates pass the filter at higher rates than infrastructure candidates do), so the drop-out leg does not corroborate the selection-bias mechanism. The relaxed-threshold leg, however, does corroborate it: the multiplier under non-selective inclusion is much smaller and not statistically distinguishable from one. The honest summary is that the published $5.7\times$ multiplier holds for the impact-screened sample but is not a robust population-level estimate; the magnitude is sensitive to candidate-pool composition, the asymmetry direction is sensitive to it, and we report the range.

5 Discussion

5.1 Interpretation of Main Findings

The $5.7\times$ infrastructure-regulatory multiplier under the baseline (impact-screened) specification, alongside the $1.5\times$ multiplier under the relaxed-threshold specification (Section 4.8), is consistent with cryptocurrency markets exhibiting differential information processing across event types within the screened sample, while not supporting

a robust population-level claim that infrastructure events are uniformly larger. Infrastructure events with multi-asset impact create immediate mechanical disruptions to trading and settlement—exchange outages halt order execution, protocol exploits drain liquidity, network congestion delays settlement—and these mechanical impairments are associated with sharp volatility spikes through direct liquidity-channel effects. Regulatory events, by contrast, operate through information channels requiring interpretation: market participants must assess compliance costs, jurisdictional implications, and long-term operational constraints, producing more gradual volatility responses on average. The selection-bias sensitivity reported in Section 4.8 indicates that this contrast holds within the impact-screened sample but does not generalise to the full candidate pool.

This distinction has theoretical significance. Cryptocurrency markets, despite continuous 24/7 trading, fragmented architecture, and retail-dominated participation, demonstrate capability to differentiate event types and calibrate responses accordingly. This challenges characterisations of crypto markets as purely sentiment-driven or informationally inefficient.

The finding that even degraded sentiment proxies improve model fit (H2) suggests cryptocurrency markets are sufficiently sentiment-driven that noise reduction is secondary to signal presence. The GDELT decomposition methodology—separating regulatory from infrastructure sentiment via article proportion weighting—represents a tractable approach for constructing thematic sentiment indices from public data. Future implementation with daily-frequency data would address temporal mismatch limitations.

The $5\times$ crisis amplification of infrastructure sensitivity (H3) reveals that mechanical disruptions and stressed market conditions interact synergistically. During crises, reduced liquidity makes markets vulnerable to operational shocks; exchange failures that cause temporary disruption in normal conditions can trigger liquidation cascades when markets are already stressed. This non-linearity has profound implications for risk models assuming linear scaling.

5.2 Network Centrality and Systemic Risk

The network spillover analysis is a supplementary observation, not a primary contribution: eigenvector centrality on a six-node correlation graph is indicative but not definitive. Within those limits, ETH exhibits higher centrality than BTC (0.89 vs 0.71, with the highest betweenness centrality at 0.42), consistent with ETH’s structural position as the foundation for DeFi protocols, NFT markets, and Layer-2 solutions. The conventional treatment of BTC as the single primary systemic risk factor may understate ETH’s role in volatility transmission, and a larger sample with more sophisticated network econometrics is a natural next step.

ETH bridges multiple ecosystem segments that BTC’s simpler architecture does not touch. Infrastructure failures affecting ETH may cascade through smart-contract dependencies, liquidity pools, and cross-chain bridges, creating multiplicative rather than additive risk propagation in our six-asset sample. The network density of 0.667 and clustering coefficient of 0.806 indicate that shocks to ETH may spread through

tightly interconnected subsystems, though the six-asset window limits the strength of any claim about broader market structure.

For portfolio managers, this suggests that traditional correlation-based diversification may underestimate systemic risk when ETH exposure is high. During infrastructure crises, ETH may act as a volatility-transmission channel, making seemingly diversified DeFi positions effectively co-exposed to ETH’s stability. Risk models that incorporate network topology metrics alongside traditional correlations are a natural extension.

5.3 Economic Significance

The statistically significant infrastructure-regulatory differential ($p = 0.0008$) translates to substantial magnitudes for portfolio risk management. Infrastructure events increase conditional volatility by 2.385 percentage points on average (ranging from 1.19% for BTC to 3.37% for ADA), representing 15–45% increases relative to baseline. This translates to annualized volatility shifts from approximately 60% baseline to 70–85% during events.

Value-at-Risk implications: Under the baseline specification, for a \$100 million cryptocurrency portfolio infrastructure events imply daily VaR increases of \$2–5 million, compared to \$0.5–1 million for regulatory events. The $5.7\times$ multiplier under impact-screened inclusion suggests that traditional “worst case scenario” planning treating all negative events as equivalent may understate infrastructure exposure within the screened sample. Under the relaxed-threshold specification (Section 4.8) the VaR differential shrinks correspondingly; the appropriate buffer ratio depends on the event-inclusion rule the risk manager adopts.

Volatility persistence: The extreme persistence parameters approaching unity suggest cryptocurrency markets operate in a near-integrated volatility regime. The half-life of volatility shocks exceeds 100 days for most assets, compared to 5–20 days in equity markets, necessitating longer hedging horizons and higher capital buffers than mean-reversion assumptions suggest.

Cross-sectional allocation: The 2.18 percentage point spread within infrastructure events (ADA to BTC) provides actionable information. ADA and ETH’s DeFi exposure creates elevated infrastructure sensitivity, while BTC’s market maturity and deep liquidity provide relative stability during operational disruptions.

5.4 Implications for Practice

5.4.1 Portfolio Risk Management

The evidence motivates differentiated hedging strategies:

- **Capital allocation:** Under the baseline specification, infrastructure events motivate $4\text{--}5\times$ higher capital buffers than regulatory events; under relaxed-threshold inclusion (Section 4.8) the differential is closer to $1.5\times$
- **Dynamic adjustment:** During identified crisis regimes, infrastructure capital requirements scale non-linearly in our regime-switching estimates (potentially several times normal buffers rather than typical $2\text{--}3\times$ stress multipliers)

- **Asset selection:** During elevated infrastructure risk, the evidence is consistent with reducing exposure to high-sensitivity DeFi assets (ADA, ETH) and increasing allocation to BTC
- **Network monitoring:** ETH’s centrality (eigenvector 0.89 vs BTC’s 0.71) suggests that seemingly diversified DeFi positions may be co-exposed to ETH stability

5.4.2 Regulatory Policy

Conditional on the impact-screened estimates that suggest infrastructure events produce larger volatility responses on average than regulatory events:

- Regulatory focus on operational resilience standards complements existing disclosure requirements
- Security auditing, disaster recovery protocols, and reserve verification appear at least as important as regulatory clarity for stability
- Circuit breaker mechanisms warrant consideration given the absence of natural cooling-off periods in 24/7 markets

5.5 Limitations

Several constraints bound interpretation:

Event classification: The infrastructure/regulatory taxonomy involves judgment calls; some events (e.g., exchange enforcement actions) have elements of both categories. Alternative classifications could yield different magnitudes.

Selection-bias and candidate-pool composition: The $5.7\times$ baseline multiplier holds in the impact-screened sample ($n = 50$) but does not survive at the same magnitude under relaxed-threshold inclusion (Section 4.8). Under a single-asset response criterion the multiplier falls to approximately $1.5\times$ and is not statistically distinguishable from one; without any impact filter the asymmetry direction inverts. The honest characterisation is that the baseline result describes the sample of events large enough to produce multi-asset spikes, not a population-level claim about all infrastructure versus all regulatory events. Reviewer 2 prompted this analysis and the result is reported here on equal footing with the baseline estimate.

Sample selection: Six major cryptocurrencies may not represent the broader ecosystem. Smaller-cap tokens, DeFi protocols, and emerging assets may exhibit different dynamics.

Sentiment data and lead-lag direction: Weekly GDELT aggregation limits temporal precision. The Granger causality tests reported in Section 4.7.7 find no evidence that sentiment leads volatility at lags 1–10; where any directional relationship is detected, returns lead sentiment. H2’s claim of incremental explanatory power survives as a contemporaneous-association statement but should not be interpreted as evidence of sentiment leading volatility. Higher-frequency (daily) sentiment data would enable sharper lead-lag resolution.

Causal identification: Event endogeneity—regulatory actions responding to market conditions—poses identification challenges. Infrastructure events offer cleaner identification as unexpected system failures (Section 3.2); regulatory events do not,

and the asymmetry of identification quality should be borne in mind when interpreting the gap between the two legs of the comparison.

Persistence dynamics: Near-integrated volatility ($\alpha + \beta \approx 0.998$) in the unconditional model appears to reflect a mixture of genuine high-persistence dynamics and un-modelled regime shifts. Bai-Perron analysis (Section 4.7.6) shows within-sub-sample persistence falling to a mean of ≈ 0.90 , suggesting the long-memory framing applies more weakly than the unconditional estimates imply.

6 Conclusion

This study presents evidence that cryptocurrency markets respond differentially to infrastructure failures and regulatory announcements, with systematically different volatility responses across our six-asset panel and across multiple specifications. Under the baseline impact-screened specification, infrastructure events are associated with $5.7\times$ larger conditional-variance responses than regulatory events (2.385% vs 0.419%, $p = 0.0008$); under relaxed-threshold inclusion (Section 4.8) the multiplier shrinks to approximately $1.5\times$, and the asymmetry collapses when no impact filter is applied. The headline magnitude is therefore conditional on the candidate-pool screening rule; we report the range honestly and emphasise the impact-screened result as the original claim while flagging the relaxed-threshold result as a substantive qualification.

Three hypotheses are examined: (H1) infrastructure events are associated with larger volatility impacts than regulatory events under the impact-screened specification, with the magnitude sensitive to candidate-pool composition; (H2) even degraded sentiment proxies improve model fit, though Granger causality tests (Section 4.7.7) indicate that sentiment does not lead returns at lags 1–10 in the weekly GDELT data; (H3) infrastructure sensitivity amplifies $5\times$ during crisis regimes in the regime-switching estimates. Two supplementary observations are noted: ETH exhibits higher correlation-network centrality than BTC in the six-asset sample, and the unconditional near-integrated volatility persistence resolves partly into a sequence of regime-conditional GARCH processes once Bai-Perron breaks are accommodated (Section 4.7.6).

The implications for practice are concrete. Differentiated capital buffers for infrastructure versus regulatory risk, dynamic scaling during identified crisis regimes, and explicit treatment of non-linear sensitivity amplification each follow from the empirical pattern documented in the impact-screened sample. The precise calibration depends on the screening rule applied to the candidate event pool, and we report a range of multiplier estimates rather than a single number for this reason.

As cryptocurrency markets mature toward institutional participation and regulatory integration, the case for understanding their information processing characteristics grows. The asymmetry direction holds under three of four candidate-pool specifications; the magnitude does not. Future work that fields a more comprehensive candidate database, daily-frequency sentiment, and a longer panel will be well-positioned to refine these estimates.

Supplementary information. Complete replication materials including data, analysis code, and figure generation scripts are available at: <https://github.com/>

studiofarzulla/crypto-event-study. An interactive dashboard is available at: <https://farzulla.org/research/crypto-event-study/>.

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- **Consent to participate:** Not applicable.
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- **Data availability:** All data are publicly available from CoinGecko API and GDELT Project. Processed datasets are archived on Zenodo (DOI: 10.5281/zenodo.17679537).
- **Code availability:** Complete replication code is available at <https://github.com/studiofarzulla/crypto-event-study> under MIT License.
- **Author contribution:** Sole author.

Appendix A Event Classification Summary

Table A1 provides selected examples from the full event dataset. Complete event list with timestamps, sources, and classification rationale is available in the replication materials.

Classification criteria: Infrastructure events affect transaction/settlement mechanics through direct operational impact. Regulatory events alter the informational environment through legal or supervisory actions. Events with elements of both categories (e.g., exchange enforcement actions) were classified by primary channel.

Appendix B GJR-GARCH-X Technical Specification

The GJR-GARCH-X model extends the GJR-GARCH(1,1) specification of Glosten et al. [22] with exogenous regressors in the conditional variance equation:

Mean equation:

$$r_t = \mu + \varepsilon_t, \quad \varepsilon_t = \sigma_t z_t, \quad z_t \sim t_\nu \quad (\text{B1})$$

Table A1 Selected Events by Category

Date	Type	Event
<i>Infrastructure Events (n=26)</i>		
2022-11-11	Infra	FTX exchange collapse
2022-05-09	Infra	Terra/Luna depegging
2023-03-10	Infra	Silergate/SVB banking crisis
2022-06-13	Infra	Celsius withdrawal halt
2020-03-12	Infra	COVID “Black Thursday”
<i>Regulatory Events (n=24)</i>		
2023-06-05	Reg	SEC sues Binance
2023-06-06	Reg	SEC sues Coinbase
2020-12-23	Reg	SEC sues Ripple
2021-05-18	Reg	China bans crypto mining
2024-01-10	Reg	Bitcoin ETF approval

Variance equation:

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \gamma \varepsilon_{t-1}^2 \mathbb{I}(\varepsilon_{t-1} < 0) + \beta \sigma_{t-1}^2 + \mathbf{X}_t' \boldsymbol{\delta} \quad (\text{B2})$$

where $\mathbf{X}_t = [D_t^{\text{infra}}, D_t^{\text{reg}}, S_t^{\text{infra}}, S_t^{\text{reg}}]'$ contains event dummies and decomposed sentiment.

Estimation: Quasi-maximum likelihood with Student- t innovations. Log-likelihood:

$$\ell(\theta) = \sum_{t=1}^T \left[\log \Gamma \left(\frac{\nu+1}{2} \right) - \log \Gamma \left(\frac{\nu}{2} \right) - \frac{1}{2} \log((\nu-2)\pi\sigma_t^2) - \frac{\nu+1}{2} \log \left(1 + \frac{\varepsilon_t^2}{(\nu-2)\sigma_t^2} \right) \right] \quad (\text{B3})$$

Constraints: Covariance stationarity enforced via $\alpha + \beta + \gamma/2 < 1$. Non-negativity: $\omega > 0$, $\alpha \geq 0$, $\beta \geq 0$, $\alpha + \gamma \geq 0$.

Coefficient interpretation: Returns expressed as percentages (multiplied by 100), so σ_t^2 is in squared percentage points. Event coefficients δ_j represent additions to squared percentage volatility during the $[-3, +3]$ day event window.

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